

Unit 4, Lesson 2: Algebraic Descriptions for Reflections



Benchmark

MA.912.GR.2.1 Given a pre-image and image, describe the transformation and represent the transformation algebraically using coordinates.



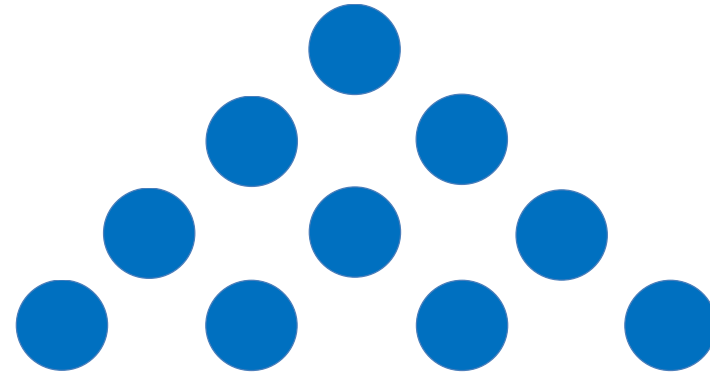
Learning Target

- ☐ I can relate a reflection on the coordinate plane to an algebraic notation of the transformation.



4.2.1 Warm-Up: Moving Circles

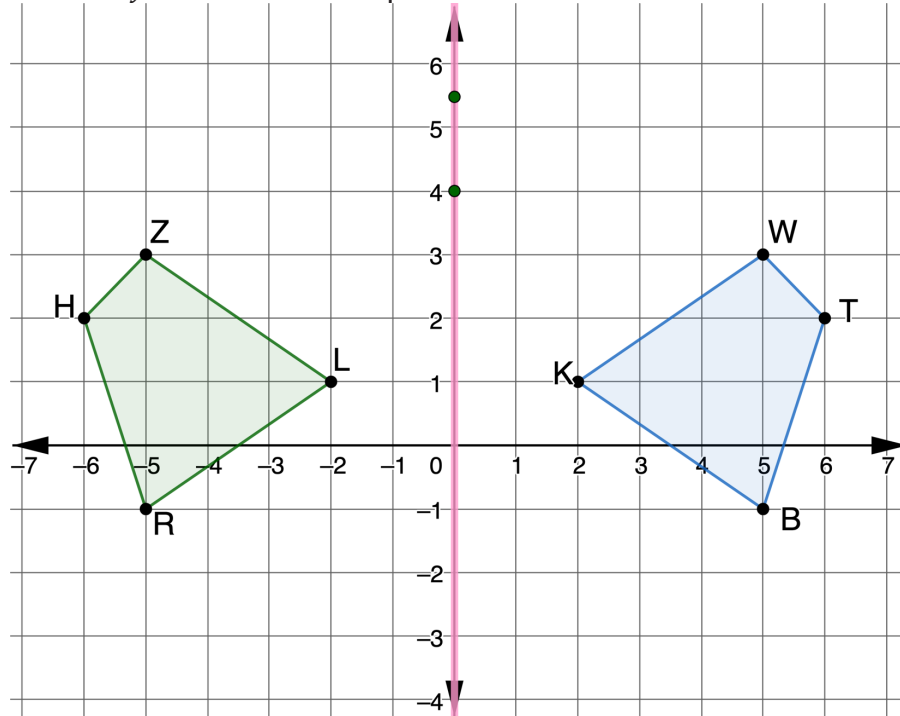
1. Move any three circles so that the pyramid points down.





4.2.2 Exploration: Discovering Algebraic Descriptions for Reflections

1. Quadrilaterals $HZLR$ and $TWKB$ are shown on the coordinate grid. Quadrilateral $HZLR$ was reflected across the y -axis to create quadrilateral $TWKB$.



- Draw a line segment between each set of corresponding vertices.
- Describe how the line segments between the vertices are related to the y -axis.

- c. Complete the table.

	H to T	Z to W	L to K	R to B
Distance				
Ordered Pair of the Midpoint				

- d. What do you notice about the midpoints?

- e. The table shows the coordinates of each vertex.

Vertex	x	y	Vertex	x	y
H	-6	2	T	6	2
Z	-5	3	W	5	3
R	-2	1	K	2	1
L	-5	-1	B	5	-1

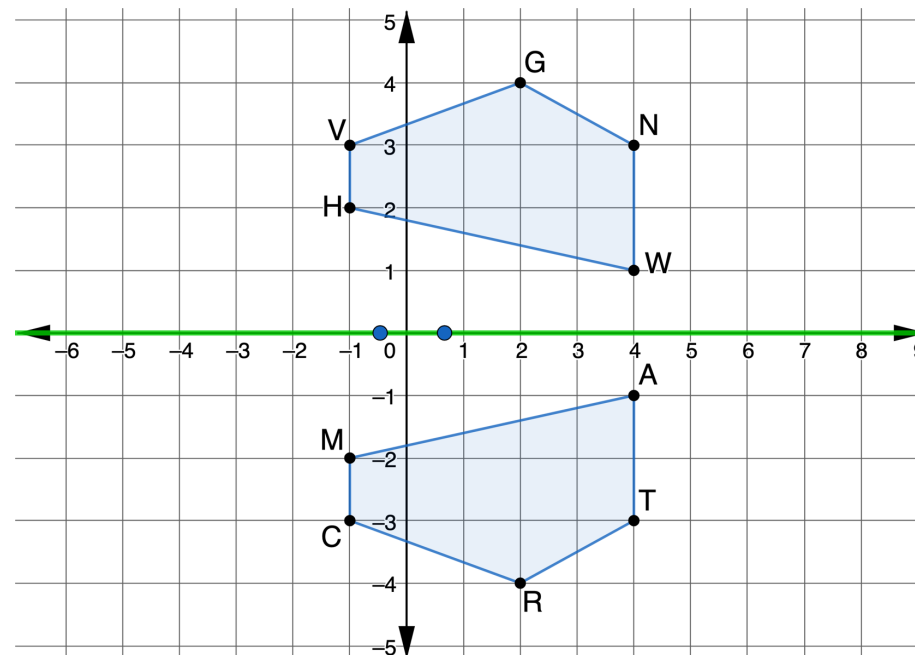
Work with your partner to describe how the coordinates of the pre-image, quadrilateral $HZLR$, are related to the coordinates of the image, quadrilateral $TWKB$.

- f. Write a conjecture that you and your partner think will work for the algebraic description of a reflection.

$(x, y) \rightarrow (\rule{1cm}{0.4pt}, \rule{1cm}{0.4pt})$

- g. Discuss with your partner if the algebraic description will work for any reflection. Summarize your conclusion.

2. Polygons $MCRTA$ and $HVGNW$ are shown on the coordinate grid. Polygon $MCRTA$ was reflected across the x -axis to create polygon $HVGNW$.



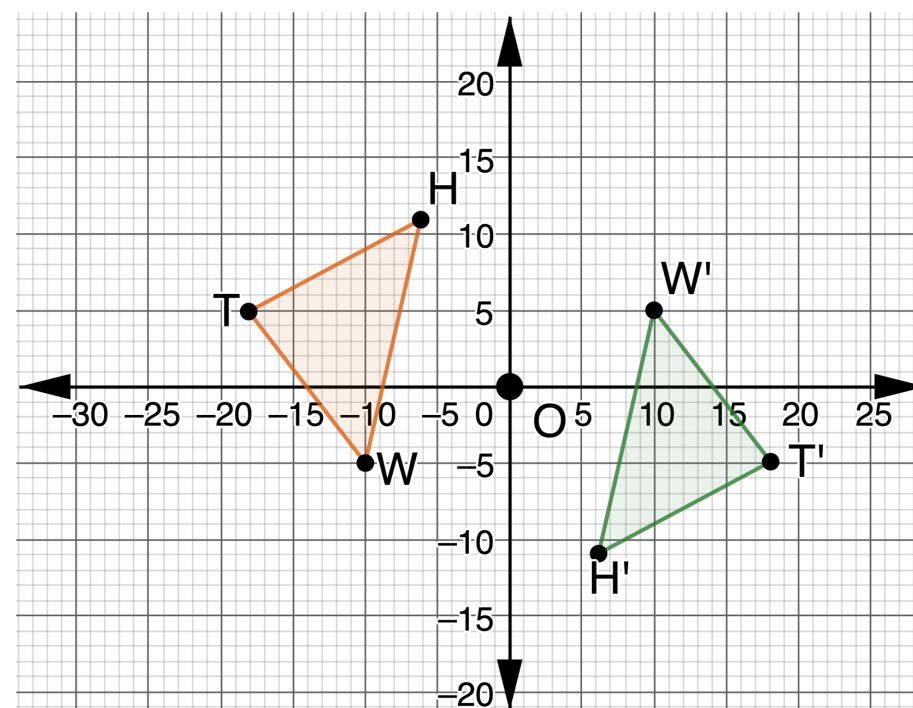
- a. Describe the relationship that the line segments between each set of corresponding vertices have with the x -axis.

- b. Wendall and Samah were working together on writing an algebraic description for reflections across the x -axis. Each student's algebraic description and explanation of their thinking is shown in the table.

Wendall	Samah
$(x, y) \rightarrow (-x, y)$	$(x, y) \rightarrow (x, -y)$
Because the polygon flipped over the x -axis, the x becomes its opposite.	Because above the x -axis all y -values are positive and below the x -axis all y -values are negative, the y becomes its opposite.

Discuss with your partner which, if any, algebraic description is correct. Use at least two ordered pairs from polygons $MCRTA$ and $HVGNW$ to defend your choice.

3. Triangles HWT and $H'W'T'$ are shown on the coordinate grid. Triangle HWT was reflected across the origin to create $\triangle H'W'T'$.

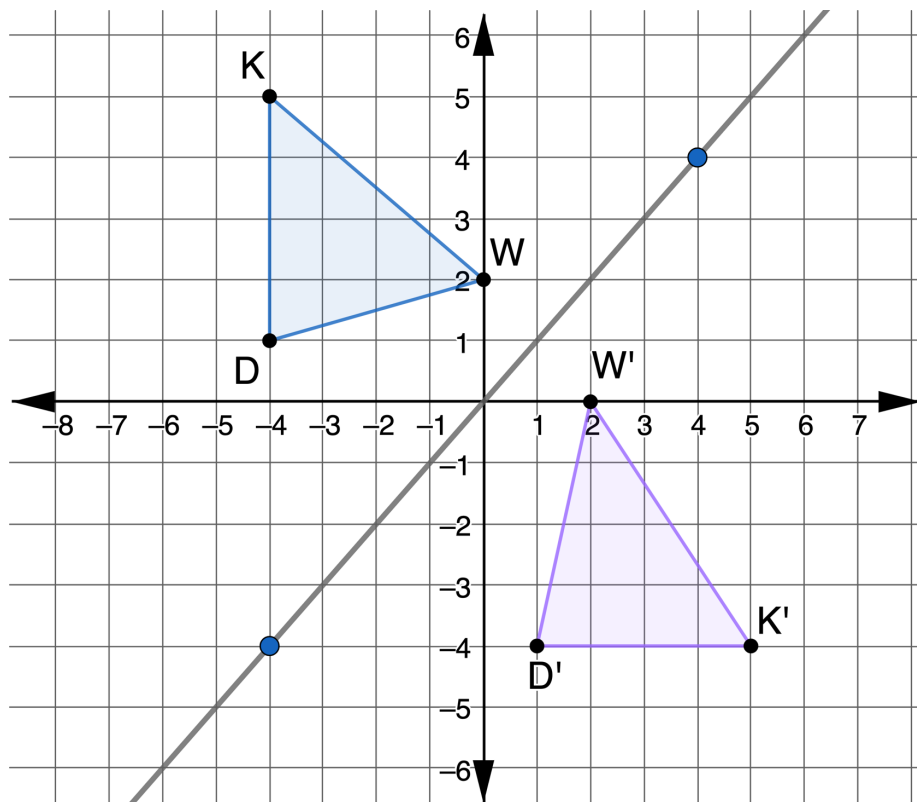


- a. Discuss with your partner how to write the algebraic description for a reflection across the origin.

$$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$$

- b. Re-examine the algebraic descriptions of rotations. Which rotation is similar to a reflection across the origin?

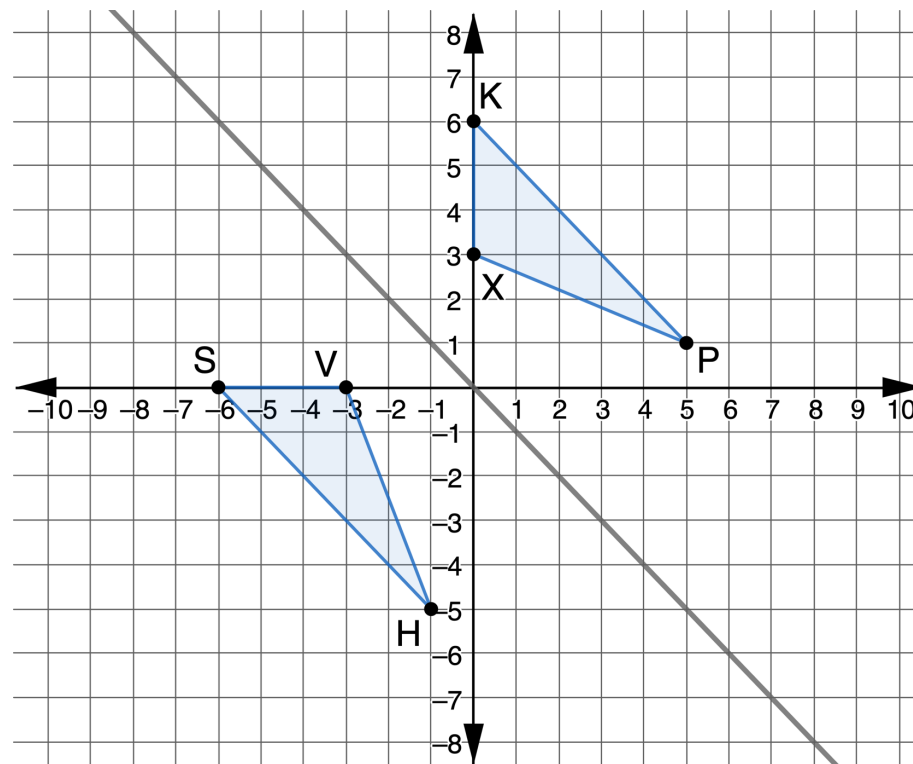
4. Triangles KWD and $K'W'D'$ are shown on the coordinate grid. Triangle KWD was reflected across the line $y = x$ to create $\triangle K'W'D'$.



Use the coordinates to determine the algebraic description for a reflection across the line $y = x$.

$$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$$

5. Triangles HSV and PKX are shown on the coordinate grid. Triangle HSV was reflected across the line $y = -x$ to create $\triangle PKX$.



Use the coordinates to determine the algebraic description for a reflection across the line $y = -x$.

$$(x, y) \rightarrow (\underline{\hspace{1cm}}, \underline{\hspace{1cm}})$$



4.2.3 Bring It Together: Algebraic Descriptions for Reflections

Reflections	
Across the x -axis	Across the y -axis
The ____-coordinate changes to its ____. $(x, y) \rightarrow (___, ___)$	The ____-coordinate changes to its ____. $(x, y) \rightarrow (___, ___)$
Across $y = x$	Across $y = -x$
The coordinates ____. $(x, y) \rightarrow (___, ___)$	The coordinates ____ and become ____. $(x, y) \rightarrow (___, ___)$



4.2.4 Wrap-Up: Cool Down

A reflection can be thought of as folding or "flipping" an object over the line of reflection. A squish painting is a fun application of a line of reflection.

To make a squish painting, a large sheet of construction paper is folded in half then opened up. Paint is dotted along one side of the fold. The paper is folded, and the folded paper is rubbed, pounded, and squished to move the paint around. The paper is then opened up to reveal the squish painting.



1. How does a squish painting show that in a reflection, each point of the original figure and its image are the same distance away from the line of reflection?